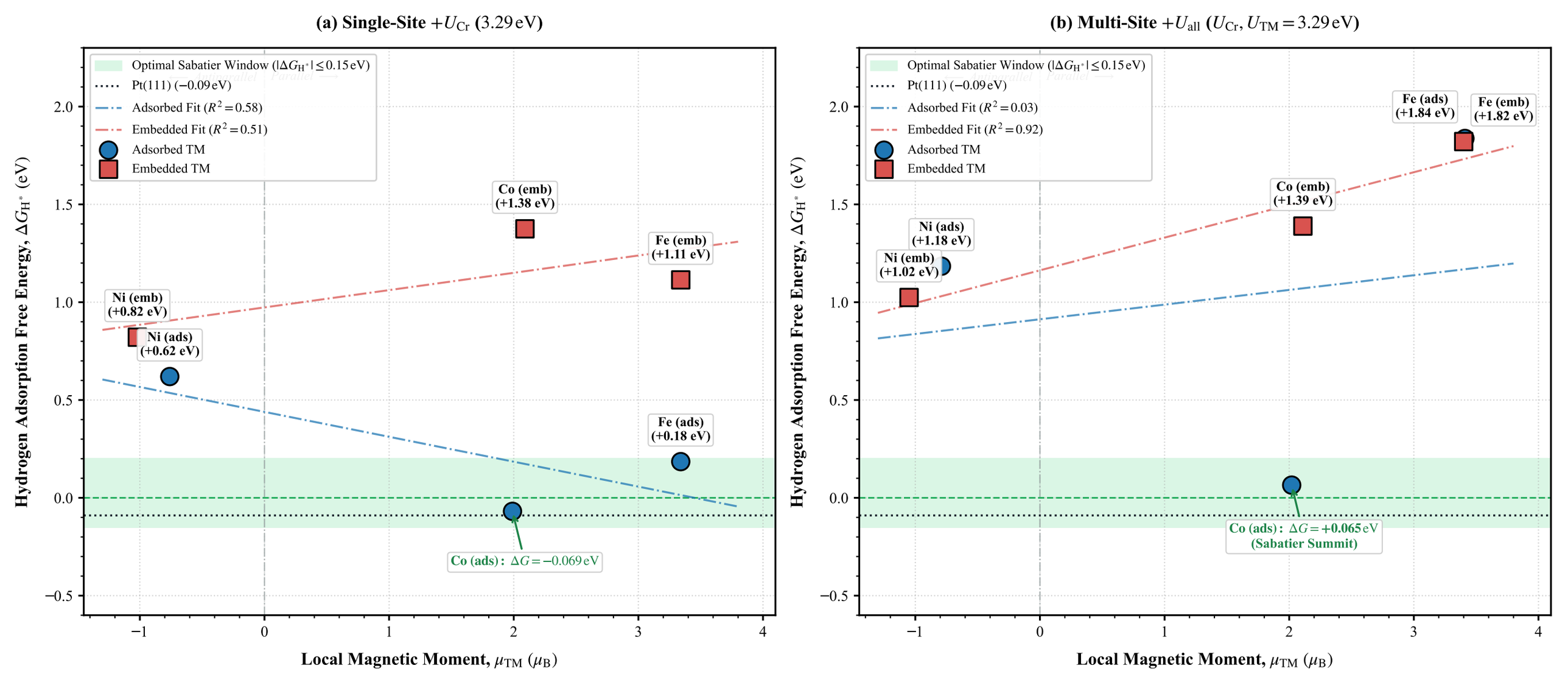

(a) Single-Site +U_{Cr} (3.29 eV)
(b) Multi-Site +U_{all} (U_{Cr}, U_{TM} = 3.29 eV)
Optimal Sabatier Window ($|\Delta G_{H^*}| \leq 0.15$ eV)
Pt(111) (−0.09 eV)
Adsorbed Fit ($R^2 = 0.58$)
Embedded Fit ($R^2 = 0.51$)
Adsorbed TM
Embedded TM
Co (emb) (+1.38 eV)
Fe (emb) (+1.11 eV)
Ni (emb) (+0.82 eV)
Ni (ads) (+0.62 eV)
Fe (ads) (+0.18 eV)
Co (ads): $\Delta G = -0.069$ eV
Hydrogen Adsorption Free Energy, ΔG_{H^*} (eV)
Local Magnetic Moment, μ_{TM} (μ_B)
Optimal Sabatier Window ($|\Delta G_{H^*}| \leq 0.15$ eV)
Pt(111) (−0.09 eV)
Adsorbed Fit ($R^2 = 0.03$)
Embedded Fit ($R^2 = 0.92$)
Adsorbed TM
Embedded TM
Fe (ads) (+1.84 eV)
Fe (emb) (+1.82 eV)
Co (emb) (+1.39 eV)
Ni (ads) (+1.18 eV)
Ni (emb) (+1.02 eV)
Co (ads): $\Delta G = +0.065$ eV (Sabatier Summit)
Hydrogen Adsorption Free Energy, ΔG_{H^*} (eV)
Local Magnetic Moment, μ_{TM} (μ_B)